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AI IN NEWSROOMS: ASSESSING PERCEPTION, CHALLENGES AND POSSIBILITIES IN BANGLADESH'S MEDIA LANDSCAPE

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**AI IN NEWSROOMS: ASSESSING PERCEPTION, CHALLENGES AND
POSSIBILITIES IN BANGLADESH'S MEDIA LANDSCAPE**

THESIS

A thesis submitted in partial fulfillment of the
requirements for the degree of Master of Arts in the
College of Communication and Information
at the University of Kentucky

By

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University of Kentucky

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Lexington, Kentucky

2024

ABSTRACT OF THESIS

AI IN NEWSROOMS: ASSESSING PERCEPTIONS, CHALLENGES AND POSSIBILITIES IN BANGLADESH'S MEDIA LANDSCAPE

With the relentless progression of technological advancements, artificial intelligence (AI) has become a transformative force across various sectors, and newsrooms are no exception. Developed nations have already integrated AI into journalistic practices, showcasing its versatility in different news functions. However, the developing nations, including Bangladesh, a south Asian nation faces challenges in catching up, marked by technological and economic disparities. As an emerging economic force Bangladesh encounters different obstacles in the adoption of AI in its media landscape. Despite the expansiveness of its media industry, the country grapples with limited technological support and inadequate training opportunities. These challenges bring up concerns regarding the familiarity and introduction of AI in mainstream journalism in Bangladesh along with the potential opportunities that lie ahead. Following the Technology Acceptance model by Davis (1989), this study aimed to assess perception among the current journalists, and editors about the usefulness and ease of AI in their daily journalism practices. Participants ($n=15$) were recruited through purposive snowball sampling from 10 different mainstream news organizations. Semi-structured questions clustered into two phases focusing on the Perceived Usefulness (PU) and Perceived Ease of Use (PEU) from the TAM in every in-depth interview. A deductive approach was followed to analyze data through thematic analysis. Finally, the findings provide the extent of familiarity, primarily positive attitude toward AI, its usefulness and ease. Additionally, how these newsrooms are considering adopting these technologies in their newsrooms, what are the factors behind the particular perceptions and roadblocks (i.e., lack of experts, training and financial allotment; language barrier, fear of new technology adaptation, and specific drawbacks of AI, such as lack of authenticity, trust, usefulness, positive aspects, etc.).

Results of this study have significant theoretical and practical implications. Theoretically, the findings provide data supporting the TAM and its two basic determinants PU and PEU by extending this work into the current perceptions of AI among journalists in Bangladesh. This research assessed the current state of Bangladeshi mainstream media, their perceptions, and the potential for integrating AI. The current

research also provides findings on the factors influencing these perceptions along with the challenges of integrating AI in future. Additionally, it provides evidence about language dominance while incorporating AI, differing perspectives between management and employees in these news organizations.

Practically, the findings of this research can be used to add literature on AI and Bangladeshi journalism as it is one of the first studies done in communication examining the integration of AI in this context. The findings can be utilized to look into the current roadblocks, emphasize on creating Bangla AI, and promote the inclusion of subjects and experts related to AI and technology in our curriculum.

KEYWORDS: Artificial Intelligence, Newsrooms, Journalism, Challenges, Media, Bangladesh

Nadia (Nahrin) Rahman

(Name of Student)

12/06/2024

Date

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CHAPTER 1. INTRODUCTION

1.1. Introduction

With the expansion of AI technology, the news organization has begun to stand out as an industry actively embracing and adjusting to the possibilities presented by AI-enabled technologies. The developed countries and their major news organizations have already adopted the broader integration of algorithms, data, and automation in journalism (e.g., Anderson, 2018; Broussard, 2019; Bucher, 2016; Diakopoulos, 2019; Lewis & Westlund, 2014a; Lewis & Westlund, 2014b; Weber & Kosterich, 2017; Lewis et al., 2019) for making media operations easier (Lewis et al., 2019). Many news organizations in developed Western countries expanded their utilization of AI technologies leading to the widespread adoption of automated news in digital journalism (Dörr, 2015; Graefe, 2016; Jamil, 2020) which has resulted in a substantial output of machine-generated news stories across diverse language and nations. These news organizations and news agencies like the British Broadcasting Corporation (BBC), The Guardian, Forbes, The Washington Post as well as the Associated Press and Reuters have been incorporating automated computers to generate weather reports, stock market updates, corporate performance, and sports coverage (Underwood, 2019; Jamil, 2020).

Now, when the USA takes the top rank in their 2021 Government AI Readiness Index, followed by the UK and Singapore, developing countries like sub-Saharan Africa, the Caribbean, and Latin America, on the other side, some central and south Asian countries scored the lowest

in this index (Chatterjee & Dethlefs, 2022). While the developed countries are focusing more on turning toward automated journalism, motivated by the need for a viable business, and to enhance their productivity in generating news content (Caswell & Dörr, 2017), the developing nations are still facing a noticeable deficiency in the integration of AI in newsrooms (Munoriyarwa et al., 2021). Several factors, including cultural, economic, and educational considerations, continue to serve as roadblocks to the adoption of AI technologies in the media sector. There is still a limited understanding of both the potential opportunities AI offers and the potential negative impacts it may have.

1.2 AI acceptance in newsrooms: A distinction between developed and developing countries

The findings of contemporary research and several reports on AI utilization in newsrooms underscore a multifaceted landscape of both positive and negative aspects. Significantly, a notable difference in perceptions became evident when comparing the viewpoints of developed and developing countries. Media organizations in developed countries are integrating AI and automated journalism to liberate journalists (Underwood, 2019) from their routine reporting, having the potential for cost-cutting, and relief from financial challenges (Fengler et al., 2021). Recent studies indicate that these newsrooms in Western Europe and North America are adopting AI increasingly in their everyday news practices (Biswal et al., 2020) which signifies a positive perception towards AI. On the other hand, developing countries and journalists there with low-income economies (DESA, 2017) face difficulties comprehending the profound changes in news-making and production processes due to the shifts brought about by AI (Graefe, 2016). Even the limited incorporation of AI in newsrooms has led to dilemmas, causing challenges for journalists (Munoriyarwa et al., 2021). They expressed their fear about the future of their jobs (Hamilton & Turner, 2021; Munoriyarwa et al., 2021), challenges in

understanding technological adaptation (Graefe, 2016), anxiety and skepticism (Van Dalen, 2012).

While AI may not bring all the possible potential to the newsrooms, it represents a valuable tool with positive sides that needs effective integration into newsrooms with a deeper understanding among the media professionals to enhance and fortify AI capabilities among them (Broussard et al., 2019, de-Lima-Santos).

1.3 The focus of this study: Bangladesh and AI in newsrooms

Bangladesh, a developing country in South Asia has been witnessing substantial growth across diverse media since the past decade (Riaz & Rahman, 2021) with a focus on making the mainstream news sector more digitalized. At present, Bangladesh has 45 private TV channels, 1248 daily newspapers, 28 FM stations, 32 community radio stations, and over 100 online news portals (Kamruzzaman, 2020; Riaz & Rahman, 2021). In this era of technological advancement, embracing digital journalism is inevitable and journalists in Bangladesh are getting equipped with essential skills; yet questions arise, merely acquiring these digital skills may prove insufficient in the technological advancement of AI (Gayen, 2020). Also, mainstream journalism is undergoing insufficient funding and budget deficits due to COVID-19 and other several factors. At present, it needs more budget and funding sources to provide constraint training on technologies to its journalists, and other employees (Gayen, 2020).

Based on the current circumstances in Bangladeshi mainstream journalism, this research examined the perceptions and understanding among working journalists and editors about AI tools in their newsrooms. Importantly, working journalists and editors are the employees of a media organization, therefore it is essential to find out their perceptions and knowledge about AI and it's

both positive and negative aspects in the current situation of journalism. Some questions to consider are: How much do they know about AI? Will it bring positive potential or negative aspects to Bangladeshi journalism? How do journalists perceive the integration of AI in their professional workflows? Prominently, how they define AI in mainstream journalism? In a developing country like Bangladesh, do news organizations have enough resources to effectively implement and manage AI integration, even if they choose to adopt these technologies in the upcoming days of technological advancement?

As a developing country, Bangladesh has been transforming into a technology-driven nation where automation and control technology are being implemented across various industries (Babu, 2021). While the adoption may have been delayed, the influence of AI, Big Data, Blockchain, and other technologies has gained popularity in different sectors there (Babu, 2021). Now it leaves the question of newsrooms and journalism practices. There is a notable gap in research addressing the opportunities and challenges of integrating AI into mainstream journalism in Bangladesh. Although research on AI has started in various areas, extensive studies that integrate both deductive and inductive approaches are still limited. Therefore, this research focused on adding new findings to the emerging literature particularly focusing on journalists' and editor's perceptions about the usefulness of AI, and the potential possibilities of integrating these technological tools in the current circumstances of mainstream journalism. Also, the research examined the underlying reasons behind their perceptions, delving into the significant perspective towards AI as well as the challenges influencing their perceived thoughts on the integration of AI in their daily professional domain.

CHAPTER 2. THEORETICAL FRAMEWORK

2.1 Theoretical Framework: Application of the Technology Acceptance Model

The technology acceptance model (TAM) by Davis in 1986 (Davis, 1989; Ma & Liu, 2008) has been followed in this study. This TAM model elucidates an individual's acceptance and overall attitude toward an emerging technology (Wan & Jiang, 2023). Here, in this model, two key determinants, perceived usefulness (PU) and perceived ease of use (PEU), will serve as fundamental factors for examining the cognition and overall perception of the participants.

PU is characterized as the extent to which an individual believes that utilizing a specific technology will improve their performance in the job, and PEU is defined as the level to which an individual perceives that working with a specific technology will require little effort compared to the bigger positive outcomes (Davis, 1989; Wan & Jiang, 2023; Zhang & Chen, 2020). This research, following the basic determinants PU and PEU, focused on how working journalists and editors perceive AI integration in improving their daily performance and credibility of the media organization. Do the employees of these Bangladeshi newsrooms think that AI integration will bring positive outcomes or there will be barriers that will impede the positive aspects?

Applying the TAM framework and specifically its two determinants, PU and PEU, this research followed the qualitative method of in-depth interviews to examine and analyze the two overarching research questions: *(RQ:1) how do the journalists and editors perceive the*

usefulness (PU) of these technological advancements of AI in their everyday practices? Also, (RQ:2) how do they perceive the ease of use (PEU) of AI in journalism practices?

CHAPTER 3. LITERATURE REVIEW

3.1 Definitions and application of AI in journalism

The conceptual definition of AI in this study is framed as the automation of primary tasks such as story creation, headline generation, translation, audio-to-text conversion, news summarization, interview transcription, and designing graphics. The beginning of automated journalism can be traced to the technological evolution of newsrooms and broadcasting over the past three decades, which has been traced with a noticeable escalation in recent years (Túñez-López et al., 2021). At present, the phrase AI in journalism encompasses algorithms and automation employed for various activities, including automated story creation, identifying news angles (Motta et al., 2020), content generation and recommendation, interview transcription, text-to-audio through synthetic voice technologies, workflow automation (Newman, 2021; Newman, 2022, Sirén-Heikel et al., 2022), fact-checking, translating (Túñez-López et al., 2020), and activities with data including data cleaning, extraction, summarization, etc. (Stray, 2019; Sirén-Heikel et al., 2022). This study, one of the first to examine the perceptions of Bangladeshi journalists and editors on AI, emphasizes addressing the gaps in existing literature. By examining the opportunities, challenges, and concerns surrounding AI in Bangladeshi newsrooms, it provides new insights into the readiness for AI adoption in journalism.

Although the newsrooms in developed countries have encompassed their applications with newsgathering, production, and news distribution in all stages of content creation, from ideation to publishing (Beckett & Yaseen, 2023), in Bangladesh, journalists have recently begun to adopt these basic functions of these AI applications, often driven by their interest than

institutional. Therefore, a fundamental research question in this study is, (RQ:3) *how do the participants define AI and its functions in newsrooms? What do they understand by AI and its tools? How much do they know about it?*

In Western developed countries already numerous technologies, categorized as AI, have been utilized in newsrooms, often labeled as automated, algorithmic, or intelligent (Sirén-Heikel et al., 2022). As definitions continue to evolve, this form of automated news text generation is also referred to as algorithmic journalism, robot journalism, automated news (Danzon-Chambaud, 2021), or even automated news production (Sirén-Heikel et al., 2019; Sirén-Heikel et al., 2022). Significantly, it is now a widely acknowledged term denoting the technology used to automate the writing of texts, a practice increasingly adopted in newsrooms (Haim & Graefe, 2017; Linden, 2016; Pashevich; 2018). Like, The Associated Press employs automation to cover 10,000 minor baseball league games each year. Beyond sports, automation is also utilized by the Associated Press to generate stories on corporate earnings (Dalen, 2012). These incorporations of AI in newsrooms within developed countries demonstrate the capability of AI software to autonomously generate news stories, interpret data, organize information, and present content in a human-readable format (Fengler et al., 2021).

The utilization of AI has created opportunities not only for major news organizations but also for smaller newsrooms with constrained resources (Hurst, 2023). Like, News Corp Australia is generating 3,000 articles weekly using generative AI. Nottinghamshire, a local newspaper in the UK uses generative AI to create concise bullet point summaries for their extended articles. AI and its tools essentially facilitate these small newsrooms to publish a multitude of local stories every week, encompassing topics such as weather updates, fuel prices, and traffic conditions (Hurst, 2023). However, in developing countries like Bangladesh, the adoption of these

technologies is still a long way off. Thus, another research question follows, (RQ:4) *do the participants think that media organizations will be able to provide technical and other resources to facilitate AI properly in the newsrooms?*

Another aspect of AI includes the implementation of intelligent or robot journalism (Tosyalı & Aytakin, 2020) which involves the utilization of software algorithms specifically crafted to generate written articles and designed to create news template. These software algorithms execute routine writing tasks traditionally performed by journalists, often formulated, and based primarily on quantitative data, exhibiting higher speed and accuracy (Tosyalı & Aytakin, 2020). In the case of the Associated Press, the adoption of automation has resulted in over a tenfold increase in the volume of earnings reports from customers (Cohen, 2015; Fengler et al., 2021). One of the prime reasons is recognizing that machines and algorithms can exhibit more comprehensiveness compared to human journalists and reporters (Underwood, 2019).

3.2 Opportunities and challenges of AI: A double-edged sword

One of the key advantages of AI-powered tools is that it facilitates newsrooms to enhance productivity and efficiency by streamlining the writing and content production process (Aissani et al., 2023). Such advanced technologies assist newsrooms in narrative creation, source identification, research, and information uncovering. Also, news organizations in developed countries integrate automated journalism as an opportunity to liberate journalists (Underwood, 2019) from routine reporting, offering them additional time for more intricate tasks.

Beyond its role in news content creation and distribution, AI has been employed to detect fake news, aiding in source verification (Feng et al., 2012; Zhang et al., 2012; Zhou et al., 2004; Rubin, 2023; Nakov, 2021). Although AI and its systems play a role in the creation of

misinformation like fake news and deepfakes, nowadays AI is effectively able to identify three distinct types of digital deception: serious journalistic fabrication, large-scale multiplatform hoaxes, and humorous fakes (Rubin et al., 2015; Rubin, 2023). Also, the source verification website, Tanibh, a news verification tool created by the Qatar Computing Research Institute aims to help both journalists and the audience uncover fake news disseminated by various Arab and global media sources (Nakov, 2021).

While AI presents positive aspects, it is also accompanied by challenges. Major challenges can be identified in algorithmic transparency (McBride et al., 2014; Dörr & Hollnbuchner, 2016), accuracy, comprehensiveness of information (Zagorulko, 2023), data usage and abuse, ethical complications of accountability while reporting (Diakopoulos, 2015; Dörr & Hollnbuchner, 2016). Bucher in his study (2018) suggests that it created a challenge in determining who would be responsible or face the challenges arising from the interaction between algorithms and human news processes (Bucher, 2018; Paik, 2023). Primary criticisms revolve around authorship such as attributing who is the author, and credibility with ensuring algorithms are impartial, accurate, and devoid of subjectivity (Gillespie, 2014; Fengler et al., 2021). Recent studies show that until now, despite the increasing trend, only a limited number of media outlets including the Associated Press (AP), the Los Angeles Times, ProPublica, and Forbes have embraced the practice of utilizing AI to rewrite content (Fengler et al., 2021).

Another challenge arises with credibility and ethical dilemmas. Graefe and Bohlken in their study (2020) shared that, overall, there was no disparity in readers' credibility perceptions. However, human-written news held a significant advantage in terms of quality, and readability. Regarding this dilemma, the BBC initiated a project, called Public Service Algorithm, which

aims to educate the audience on algorithm usage and to guide them on how to appropriately customize the use of applications (Túñez-Lopez et al., 2019).

Consequently, at the local news level, journalists are facing concerns with traditional ethical guidelines (Paik, 2023). Many of these guidelines are becoming obsolete and unattainable due to the increasing influence of algorithm systems integrated into the current news-making process.

The application of algorithms in professional news-making processes gives rise to ethical challenges in various contexts (Dörr & Hollnbuchner, 2016). Bradshaw (2019) argued that ethical dilemmas emerge at the level of data search and its origin, while Carlson (2014) suggested the specific authority of algorithms. Gillespie (2014) indicated concerns about objectivity within these algorithms (Dörr & Hollnbuchner, 2016). One major apprehension revolves around the collection of user information and the categorization of users based on similar information (Cao, 2023).

Some fundamental elements of news are emotion and empathy (Plaisance, 2019) which the AI lacks (Cao, 2023). While generating news that demands emotional appeal, AI is unable to grasp the emotional nuances of humans, or even to sense the social temperature with a lack of resonating with people (Cao, 2023). This is exemplified by the AI news presenter Aparajita, launched by Channel 24 in Bangladesh. While the technology reads and presents content in a tech-based program it lacks pace, coherence, and importantly emotional appeal. Participants in this study from other news organizations shared their views, reflecting on audience perspectives and their perceptions of this technology. Based on this research question follows, (RQ: 4) *how do participants perceive the role of human journalists versus AI in journalism? Will be there any competition and conflict between AI in journalism and human journalists?*

Another dilemma emerges regarding control over news dissemination, pitting journalists against algorithms (Paik, 2023). The widespread dissemination of news and content implies reduced control for journalists and editors over content distribution to consumers. On one hand, algorithmic systems deliver personalized and tailored news to users; but on the other, they confine users in an "information cocoon room," by limiting users' openness to contrary information by undermining inclusiveness (Cao, 2023). Thus, based on these challenges and uncertainties, a research question follows (RQ: 5) *To what extent do journalists and editors in Bangladesh place their trust in AI technologies?*

3.3 AI acceptance in journalism: TAM perspective

Recent studies indicate a growing adoption of AI in daily news practices in Western Europe and North America (Biswal et al., 2020). On the other hand, the use of AI in news-making practices is still relatively new in developing countries (Jamil, 2020). A study from the United Nations Department of Economic and Social Affairs (DESA, 2017) indicates that there is a limited understanding of the possible impact of AI on different sectors in developing nations with low economies. These factors collectively play a significant role in shaping the perceptions of newsroom employees towards technological integration, significantly AI, which is still a new concept in developing countries like Bangladesh. Furthermore, it is noteworthy that the majority of research within the discipline of AI, specifically in media and journalism, is primarily conducted in developed countries with large economies including the United States, the European Union, Scandinavia, and China (Jamil, 2020). Even the limited incorporation of AI in newsrooms of some developing countries has led to dilemmas, causing challenges for their journalists (Munoriyarwa et al., 2021). Thus, another research question in this study follows,

(RQ: 5) what are the roadblocks to integrating AI into journalism in a developing country like Bangladesh?

The TAM was developed later by incorporating additional determinants under PU (Perceived Usefulness) and PEU (Perceived Ease of Usefulness). In TAM3 (Venkatesh & Bala, 2008; Wan & Jiang, 2023) six additional determinants were incorporated in PEU including self-efficacy in computer use, external control perceptions, computer-related apprehension, playful engagement with computers, perceived enjoyment, and perceived usability of the system. These additional determinants, which address the user's specific beliefs and attitudes, are connected to various cultural, educational, and economic factors. This is also relevant to the concept of technology appropriation which assesses how individuals within an organization adopt as well as adapt to a specific technological system (Janneck, 2009; Munoriyarwa et al., 2021). This concept is significantly related to the organization's overall political, economic, and social contexts, and importantly the working practices (Janneck, 2009; Munoriyarwa et al., 2021). Collectively, these factors contribute to forming a user's overall perception.

In the newsrooms of developing countries, where organizational and technical resources are insufficient, this situation creates uncertainty among journalists and other employees. In the study on AI and the perception of Pakistani journalists, Jamil (2020) showed that developing countries, with their unstable economy, and challenging political and technological environments, are still in the early stages of integrating AI-enhanced journalistic practices. The interviewed journalists shared their perceptions about uncertainties, particularly regarding the lack of economic and technological resources, as well as the deficiencies in education and training related to artificial intelligence and digital journalism (Jamil, 2020). South African

journalists and editors shared a pessimistic attitude due to factors such as a shortage of skills and insufficient funds within newsrooms (Munoriyarwa et al., 2021).

The determinant under PEU is Computer-related apprehension or anxiety which is an individual's fear when he or she confronts the prospect of utilizing computers or new technology (Venkatesh et al., 2003; Venkatesh & Bala, 2008). On the other hand, another determinant of computer playfulness is the level of cognitive spontaneity in interactions with computers (Venkatesh, 2000; Venkatesh & Bala, 2008). Based on these two determinants, the attitudes of journalists and editors can be addressed in two distinct ways. First, they may anticipate technological apprehension. On the contrary, they may perceive facilitation and spontaneity resulting from the integration of AI in newsrooms. A notable concern among journalists is the apprehension about the future of their employment, reflecting a fear that artificial technologies and algorithms might replace them in their roles (Hamilton & Turner, 2021; Munoriyarwa et al., 2021).

These fears and concerns pose significant challenges in those developing countries (Biswal et al., 2020; Munoriyarwa et al., 2021). Van Dalen (2012) noted that journalists expressed their mixed feelings including anxiety and skepticism resulting from the implementation of AI. Similarly, Weaver (2020) emphasized the impact of AI on the declining number of full-time journalists, as artificial intelligence and automated technology are taking over their jobs (Munoriyarwa et al., 2021). Therefore another (RQ: 6a) focuses on, *how the participants view AI integration. Is it characterized more by fear or apprehension, or is there a sense of playfulness associated with it?*

One of the reasons behind the scenario, according to cognitive psychologists, individuals are notably tend to fear the unknown and unfamiliar (Brockner et al., 1987; Wanberg & Banas,

2000; Jung et al., 2017). Another factor to consider is a culture that creates a perception among journalists, which can be elaborated with one of the additional determinants, specifically job relevance. In the TAM2, under PU the job relevance (Venkatesh & Davis, 2000; Wan & Jiang, 2023) refers to an individual's perception of the extent to which the target system is relevant to their job. Many journalists in developing countries hold a pessimistic view regarding the integration of AI, addressing it as "a Western practice." Culture is one of the important factors that holds a strong position in establishing and maintaining trust (Linden, 2019; Yu & Huang, 2021). A study in South Korea showed that audiences and journalists assigned higher ratings to the content produced and works completed by AI over that of a human journalist (Jung et al., 2017; Yu & Huang, 2021). The reason behind this negative perception among the public and media employees was unethical practices within the media organization (Humphreys, 2005; (Yousefikhah, 2017).

Thus, a research question follows, (RQ: 6b) *Are there any cultural or technological roadblocks behind the apprehension regarding AI? If so, what are the primary concerns identified by the participants?*

While AI may not bring all the possible potential in the newsrooms and era of journalism, it represents a valuable tool that needs effective integration with a deeper understanding among journalists and media professionals to enhance and fortify the technological capabilities (Broussard et al., 2019, de-Lima-Santos). A contrary picture is evident in developed countries like scholars in media and journalism have acknowledged the emerging application of AI in their news-making and other functional processes (Diakopoulos, 2019; Lewis et al., 2018; Lewis et al., 2019; Caswell & Dörr, 2017; Carlson, 2016; Smith & Eckroth, 2017; Lindén, 2017; Linden,

2016; Jamil, 2020). Consequently, there has been a rise in the studies and research fields to examine the utilization and significance of AI within the context of journalism.

Several newsrooms in developing countries also have highlighted some positive aspects of integrating AI into their operations. Bloomberg, one of the South African multiplatform news organizations (Machin & Niblock, 2010; Munoriyarwa et al., 2021) integrated AI which helped the organization in modifying headlines, notably, in the rapid production of breaking news (Munoriyarwa et al., 2021). Bloomberg has implemented a variety of software to update their financial news and monitor readership which made news production an easier process. Although journalists in this study shared their positive views on the usefulness of AI, they indicated financial constraints and insufficient technological support as key challenges, hindering the adoption of AI practices in their newsrooms (Munoriyarwa et al., 2021).

Overall, the integration of new technology is an outcome shaped by economic, social, and cultural factors. Additionally, it is shaped by human perceptions, attitudes, and actions. In Bangladesh, AI integration in journalism has not started yet. Limited implementations including AI-generated presenters for news presentations is observed in only one TV channel, Channel 24. Therefore, it is important to examine the perceptions of journalists and editors about this technology, considering that they are the professionals who will be dealing with AI and its tools in their newsrooms.

CHAPTER 4. APPLICATION OF TECHNOLOGY ACCEPTANCE MODEL (TAM)

4.1 Application of TAM model as theoretical framework

This study examined journalists' and editors' perceptions of AI by significantly following the additional determinants associated with the fundamental factors of Perceived Usefulness (PU) and Perceived Ease of Use (PEU) in TAM. In the TAM2, five specific additional determinants were added under the PU including subjective norm, image, job significance (mentioned before), output excellence, and demonstrability of results (Venkatesh & Davis, 2000; Wan & Jiang, 2023). The determinants, excellence, and demonstrability of results are defined as an individual's perceptions of a system's task performance capabilities (Venkatesh & Davis, 2000; Wan & Jiang, 2023) and the concrete outcomes derived from the utilization of the innovation (Moore & Benbasat, 1991; Wan & Jiang, 2023). Like, what does a journalist think about AI's work performance? Does s/he think the AI is credible meeting the required standards expertise in newsrooms? What is the participants' perception of AI in bringing out concrete outcomes as a result? All these questions fundamentally align with the main research question (RQ:1) of this study.

Other six additional determinants were incorporated in PEU including self-efficacy in computer use, external control perceptions (described above, in the 3.3 'AI acceptance in journalism: TAM perspective'), computer-related apprehension, playful engagement with computers, perceived enjoyment, and perceived usability of the system in the TAM3 model (Venkatesh & Bala, 2008; Wan & Jiang, 2023). Five determinants from this TAM3 are aligned with the aims of this study.

Here self-efficacy in computer use is defined as the extent to which an individual perceives their capability to carry out a particular task utilizing a computer (Compeau & Higgins, 1995a; Compeau & Higgins, 1995b; Venkatesh & Bala, 2008) or any new technology including AI and algorithmic systems in newsrooms. And, this research examined, how journalists and working editors perceive their ability to utilize these AI-enabled systems. This aligns with the (RQ:3) which focuses on, how much the participants know about AI and how they are defining it.

The fourth and fifth determinants computer playfulness and perceived enjoyment both mean the level of cognitive spontaneity in interactions (Venkatesh, 2000; Venkatesh & Bala, 2008) and playfulness (Webster & Martocchio, 1992; Venkatesh & Bala, 2008) with computers. This study examines apprehension and concurrent playfulness regarding AI in newsrooms through in-depth interviews, focusing on whether participants anticipate technological apprehension or perceive AI as facilitating spontaneity and ease if implemented.

The reasons for applying the TAM model in this study are two-fold. First, the particular determinants in the model helped to assess the perceptions of the participants about the integration of AI in their daily work. It also helped to understand, how these participants perceive the relevance of AI to their daily work.

Some additional variables in the very first TAM including trust and knowledge were added to bolster this model's predictive efficacy (Kashive et al., 2020; Lin & Xu, 2021; Kelly et al., 2023). Here, trust plays a pivotal role and is incorporated into the technology acceptance model to assess individuals' behavioral intentions (Kelly et al., 2023). Expanding the TAM model in 2022, Choung et al. identified a significant and positive correlation between trust and perceived usefulness (Choung et al., 2022; Kelly et al., 2023). Trust in AI and technology

providers act as a driving factor for AI acceptance among journalists (Kelly et al., 2023). The integration of AI in newsrooms may be reinforced if employees maintain trust in this technology, perceiving it as genuinely aiding them in accomplishing tasks more efficiently and effectively.

Summarily, evaluating user acceptance is essential for stakeholders in any organization to comprehend the necessity to optimize the adoption of technology (Kelly et al., 2023). Thus, in this qualitative study, TAM is followed to develop the semi-structured questions for the in-depth interviews to examine the nuanced perceptions of the participants regarding the integration of AI in newsrooms, the roadblocks, and possible positive aspects. The questions aimed to explore how participants thought AI tools would be easy to use and whether they believed these tools would help improve efficiency, content production, or decision-making in their work. Additionally, particular determinants under the PU and PEU were followed into the interview questions to examine, how journalists feel about working with AI, and whether they viewed it as an enjoyable tool or something primarily out of necessity.

4.2 The case of Bangladeshi journalism and the possibility of AI in newsrooms

Bangladesh, a democratic country and an emerging economic power after its independence in 1971, it experienced significant expansion across various media channels over the past decade (Riaz & Rahman, 2021). It has been ranked 162 out of 180 in the Press Freedom Index (2023) according to the World Press Freedom Index for 2022, assembled by Reporters Without Borders. Article 39 of the country's constitution guarantees the freedom of thought, conscience, and the freedom of the press (Government of Bangladesh, 1972). Since its independence, Bangladesh has undergone both military and non-military regimes (Islam, 1970). Despite enduring different turmoil including political instability, natural disasters and calamities, overpopulation, poverty, and low literacy rate, this country has experienced a turnaround and

identified itself as a “development paradox” for its changed economic growth, specifically in recent decades (Riaz & Rahman, 2021).

Though there is limited availability of data and information concerning the economic size of the media in Bangladesh (Riaz & Rahman, 2021), an online report citing corporate research shows that the media industry in Bangladesh was valued at 27 billion BDT in 2016 (Ghosh, 2017; Riaz & Rahman, 2021), exhibiting an annual growth rate of around 10-12 percent. This report also indicates that the television industry has seen substantial growth of 200 percent, print media by 150 percent, radio by 350 percent, and digital media by an impressive 1,500 percent during the 2012-2016 period.

Nevertheless, when the financial dilemma poses challenges for newsroom professionals seeking basic salaries and other increments (Mondol & Paul, 2021); it is a pertinent question of the eligibility among the news organizations to integrate AI. This is related to the perceived usability of the system, a key determinant of PEU, which posits that factors such as cultural and economic influences shape an individual's attitudes toward adopting new technology. Mainstream journalism has been undergoing insufficient funding and budget deficits (Gayen, 2020). The primary income source for the newspapers is advertising and this income has decreased since the onset of the pandemic. What adds to the drawback is that a growing share of the constrained advertising budget is flowing toward social media content that may not necessarily align with journalistic principles (Gayen, 2020). As an example, a significant amount of telecommunications funding, for instance, is being funded for Wazz-related content (religious preaching in Islam) on social media which has never been categorized as news in mainstream journalism. However, due to its ability to attract more viewers, ad companies are allocating advertisements for this content. Also, newsrooms in Bangladesh, particularly most of the

privately owned television channels, face frequent criticism for their absence of specific ethical guidelines and policies for the integration of AI (Gayen, 2020).

Significantly, the media landscape has been negatively affected by the Covid-19 pandemic (Saif, 2020), and journalists' and other media professionals' salaries were delayed during the whole pandemic (Saif, 2020) and even after. The Dhaka Union of Journalists verifies that approximately 1000 journalists either faced job losses or encountered salary non-payment amid the pandemic, as the country was impacted by Covid-19 (Riaz & Rahman, 2021). Professional forums, including the Newspaper Owners' Association of Bangladesh (NOAB), the Association of Television Channel Owners (ATCO), the Organization of Newspaper Editors, Bangladesh Sangbadpatra Parishad, and Dhaka Reporters Unity, demanded protection from the government without achieving any positive response or success (Sharifuzzaman, 2020).

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On the contrary, journalists in Western as well as developed countries along with China, are increasingly integrating AI into their newsrooms, which is transforming the landscape of

their journalism (Kothari & Cruikshank, 2021). In their study, Yu and Hang (2021) conducted interviews with Chinese journalists where they revealed their positive perception of AI as a tool that “liberates human journalists” (Yu & Huang, 2021; Kothari & Cruikshank, 2021).

Conversely, journalists in the developing countries mostly shared their pessimistic attitudes and perceptions about AI as a system of “Western culture”, shared their anxiety about AI which can exclude them from their jobs, and questioned its efficiency in their newsroom tasks (Jamil, 2020; Biswal et al., 2020; Munoriyarwa et al., 2021). Cultural perceptions about AI like skepticism, mistrust, and Western ideology influence the acceptance of AI, especially if journalists perceive it as a threat to job security. Concerns over AI replacing human creativity, and control over content can also foster resistance, particularly among the journalists who have technological apprehension. On the other hand, the younger generation of journalists might be open to AI, viewing it as a tool to enhance efficiency.

CHAPTER 5. METHODOLOGY

In-depth interviews were conducted with current working editors and journalists as participants, primarily sampled from ten different mainstream news organizations. The researcher herself collected the data right after getting the approval from the IRB, University of Kentucky. Initially, the researcher communicated to the participants including editors, and journalists through e-mails, and utilizing other communication platforms like WhatsApp and Messenger to schedule the interview. Interviews were conducted until data saturation was reached in this qualitative study.

5.1 Recruiting Participants and Sampling

Participants in this study included senior reporters, reporters, special correspondents, executive directors, newsroom editors, and sub-editors. Additionally, editors with different designations from various beats, such as joint-news editors, political editor, and the founder of an online news portal, were also recruited. These journalists and editors have a wide range of experience, spanning from five to over twenty-five years in journalism. For instance, the participants from Samakal and Channel24 have more than nineteen and twenty-five years of experience, while another participant from Somoy TV has five years of experience. The diversity in participant experiences, roles, and media sectors enriches the study with varied perspectives and strengthens the reliability of findings.

The study used purposive sampling to select mainstream news organizations that stand out in terms of highest circulation and audience reach. The inclusion of both purposive and snowball sampling added depth to the participant pool. Newspapers including Daily Prothom Alo, and The

Daily Star, The Daily Janakantha, The Daily Ittefaq get printed more than 2,85,000 copies each day and have around 6.9 million visitors (BDINFO.COM, 2023). Primarily the sampling pool was thought to consist of five newspapers, five television channels, and five online news portals that have successfully reached a maximum circulation and audience over the past five years. As the current study concentrates on the incorporation of AI into the broader media landscape in Bangladesh, it will help to gain contemporary insights through the involvement of diverse media entities, encompassing print, broadcast, and recently emerging online platforms.

Later, during the interview phase, several selected participants through purposive snowball sampling, were unable to attend the final sessions. Ultimately, interviews were conducted with seven participants from five television news channels (including one editor working in the social media segment of the TV channel), four participants from two newspapers, and four participants from three different online news portals.

Finally, the selected media organizations are:

	Name of the Media Organization	Number of The Recruited Participant
Television Channel	Jamuna TV	2
	Channel 24	2
	Somoy TV	1

	Dhaka Bangla Channel (DBC)	1
	National Television (NTV)	1
Online News portal	Aparajeo Bangla	1
	Bangla Tribune	2
	Dhaka Post	1
Newspaper	Prothom Alo	3
	Samakal	1
Total participants: $n=15$		

Among the selected media channels, Channel24 has been the first to implement an AI female presenter, Aparajita in their news broadcasts. While there have been criticisms regarding the extent of real AI involvement in the news production process, Channel24 has inaugurated this technology among all the media in Bangladesh, thus this broadcast channel is selected. The focus is on understanding the current state of their AI technology and their future development plans.

While recruiting the participants, their experience working tenure in this profession, and efficiency in the digital technological arena are prioritized. Editors in a media organization know the significant policies while keeping the information with a deeper understanding. Similarly, journalists who have insights about the particular newsrooms and have more working experience in this field have been prioritized for recruitment. Specifically, two special correspondents (one of whom is also the shift-in-charge of the digital and feature department), two senior reporters, one reporter, one senior editor, four editors (including the founder of an online news portal), one newsroom editor, one executive director, one head of news video, and one journalist designated as the head of research and planning were recruited. No additional exclusion criteria were applied. Their collective experience in journalism represents more than 203 years, and the sample of participants ranged in age from 26 to 57 ($M = 41$, $n = 13$ male, $n = 2$ female).

Among these journalists, many also have experience working as news presenters and anchors on talk shows. One positive aspect among these respondents is that each of them has experience working in various media organizations, including newspapers, TV channels, and online news portals. These participants developed a comprehensive understanding through their extensive professional involvement and expertise.

Also, it was tried to maintain a balanced gender ratio among the participants, but this was not achieved during the final interviews. Additionally, it was one of the challenges to find female journalists in senior positions, such as executive director, due to their less representation compared to male employees. This study does not include participants from rural newspapers or local correspondents. It focuses on major news organizations based in Dhaka, chosen for their large circulation and audience reach. Another reason was the challenge of reaching participants, as

interviews were done on Zoom while fast internet connections were not always available in local areas.

After obtaining IRB approval and after obtaining consent from the participants, semi-structured one-on-one interviews were conducted using the online Zoom platform, due to the time convenience both for the researcher and the participants. Each interview was for around 40 minutes to more than 60 minutes and every interview was recorded. Every participant was provided the informed consent documents ahead of time via e-mail and at the start of the interview session, gave verbal consent.

5.2 Exclusion Criteria

All participants were selected from mainstream news organizations based in Dhaka, as these news organizations are more likely to integrate advanced technology-related training and workshops for their employees. Also, they have organized policies and guidelines for their news channel. Although participants only shared their limited and very basic understanding of AI. This limited knowledge also extends to local news organizations, which is why they were not included in a later phase, as it was unlikely that significant data would emerge from these local news organizations.

Radio news organizations were also excluded because, in Bangladesh, the 25 privately owned FM stations mainly focus on entertainment. Although Bangladesh Betar, Bangladesh Betar Traffic Broadcasting, these government-run stations provide mainstream news content, none of these have yet integrated AI.

5.3 Interview Analysis

For consistency, the same interview guidelines and questionnaires have been followed in every interview. Open-ended questions were asked during semi-structured interviews. Participants were asked questions segmented into two clusters. The first cluster included the questions relevant to their perceptions about AI, do they trust these updated technologies, what and how much they know about AI, if they feel comfortable or have any concerns (i.e., losing jobs, having lower salaries), what would be the possible benefits of applying AI in their newsrooms, etc. Applying the TAM model, on the flip side, the second cluster focused on the reasons behind their perception including current obstacles that hinder the incorporation of AI-enabled devices, and automated journalism in their work environments. If they have a favorable perspective on AI, what are the key reasons behind their positive outlook? In some interviews, follow-up questions were asked to acquire a comprehensive understanding of newsrooms and their perspectives on AI integration, aiming to grasp the overall insights and scenario.

As all the participants are from Bangladesh, the interview sessions with the research questions and data collection also followed in Bengali (Mother language). After fifteen one-to-one interviews were completed, Zoom recordings were translated into English by using a free software, OnlineDocTranslator. Before the translation, every interview was transcribed automatically in the Zoom platform while conducting the interviews. While transcribing at Zoom, audio recordings including incomplete words, restarting phrases, and words, redundancies, as well as different emotional expressions, everything of these conversations were transcribed. Translated transcripts totaling 382 single-spaced pages of text, were analyzed following the thematic approach. The researcher herself went through the transcripts several times following the inductive techniques (Lindlof and Taylor, 2011; Spencer et al., 2019) and the

primary two overarching research questions. In the process of reviewing the translated transcripts, the researcher carefully examined any potential changes in meaning or context due to translation. Any discrepancies or inaccuracies in meaning were manually corrected by the researcher to ensure consistency and clarity. Each participant is given a code number due to privacy and security concerns. The code numbers (i.e. participant:01) are assigned according to their alphabetical order of names.

5.4 Data Analysis

To analyze data, thematic analysis (Braun and Clarke, 2009) is followed where themes emerging from the data are used. As a distinct research method in itself (e.g., Braun & Clarke, 2006; King, 2004; Lininger, 1992; Thorne, 2000; Nowell et al., 2017), this approach proves to be a valuable approach to exploring the viewpoints of diverse research participants, emphasizing both commonalities and disparities and yielding unforeseen insights (King, 2004; Nowell et al., 2017). Applied in this study, this approach helped to assess the different viewpoints of research participants including their perceptions, different opinions, and the reasons behind these factors. This approach also helped to clarify the participants' viewpoints as well as the factors to shape their point of view (Jason et al., 2016). Also, this study has followed a deductive approach from the outset. While developing the themes, the research questions were initially guided by a deductive approach. Later, an inductive approach was also adopted as additional data emerged during the interviews.

Conducting the thematic analysis started with the identification of themes within the dataset of in-depth interviews. Next, the researcher herself organized the data into specific themes. Further, identifying secondary themes within the data, allowed for greater interpretation (Braun and Clarke, 2006; Jason et al., 2016) including the participants' particular perceptions

and specific opinions about the integration of AI in newsrooms. Here, open coding is followed to label similar meanings and assign specific meanings that align with the aims of this study. Then axial coding is followed to link and organize the code categories.

CHAPTER 6. FINDINGS

The findings of this qualitative analysis suggest an experimental, limited but positive uptake of AI practices in the mainstream newsrooms of Bangladesh. Each respondent admitted their utilization of particular AI tools on a personal level out of personal interest, but this usage remains still low. Secondly, every participant expressed optimism about AI and shared their positive perceptions, stating that it is infeasible to avoid these technological tools in future journalism even in a developing country like Bangladesh. They highlighted the different waves of technology, noting that we are currently in the second wave of digitalization while maintaining a welcoming mindset for the third wave, which involves AI. Out of the fifteen respondents, only two conveyed direct apprehension regarding potential job losses. First, this is because of the significant factors of the present financial situation, job losses, constrained budget, and the absence of proper policies and guidelines in their newsrooms.

All respondents contended that their media organizations lack sufficient technological expertise to provide adequate training on AI and its tools for the news-making process. Second, there is a lack of proper understanding of AI and how it can be utilized in daily tasks within newsrooms. Finally, a crucial aspect of this analysis is the absence of Bangla AI, which creates significant challenges for journalists, hindering their efficiency in tasks such as translation, transcription, and accessing reliable information on topics related to Bengali literature, history, and politics. It also indicates a subtle dominance of particular languages even in this globalized era. Combining all these factors presents specific challenges in integrating AI effectively. However, this finding suggests an overarching outlook among the respondents is optimistic, with

three of the sampled media organizations already started their initiative of AI integration through the inclusion of technological experts, planning, and pilot projects.

6.1 Emerged Themes

Based on the TAM model, this analysis focused on the two overarching research questions examining participants' perceptions of the AI's usefulness and its ease of use in their current newsrooms. Fundamental themes emerged in this analysis phase: *very limited, personal, and experimental use; Absence of ease, A double-edged sword, Dual perspectives, Roadblocks to AI, Potentials, and possibilities of future application.*

Table 1. Exemplar Participants Quotations

Main Themes	Sub-themes	Participants Exemplar Quotations
Perceived usefulness and the early stages: Very limited, personal, and experimental use	In experimental basis	‘Online and Television media are using AI in an experimental basis also for some time, but it is not fully started from organizational level. Journalists are also using AI in their newsrooms in their own personal interest.’ ‘I did not use that much for organizational works, but I like to explore personally, like sometimes for writing, checking information to get an idea.’
	Selective applications	‘For now, the AI-generated presenter is only doing voice-over. We have a dedicated producer for this task basically. There is also an editor for the program who is in the charge of control of producing this program and he prepares the content by himself. After preparing the content with information, scripts in Bengali, he finally uploads it to a software. Then AI presenter whose

		name we have given as Aparajita, reads the news or present the information.’ ‘Well, here is misconception again. How would you define news making process? Yes, Channel24 has done a big branding by integrating AI presenter in their news presentation, but there are man forces who are working behind this. It does not work like a basic AI tool where it creates content itself and distribute by itself too. Rather it lacks any kind of emotion, tones that a human presenter would have.’
Absence of Ease	Still at theoretical level	‘Because the topic of artificial intelligence is new in Bangladeshi newsrooms and its not that long we have started talking about this, so those of us who actually work professionally are still at the theoretical level. Practically, it has not yet been possible to work because it has not reached at a very good professional level.’ ‘Actually, the matter is very new in Bangladesh, I can say that it is not that familiar. This is still very, very rare and has been used as a private initiative. If we talk about all the journalism in Bangladesh, then only one thing I can say that it is not launched officially.’
	AI as time savior	‘When I was working with it, it would take maybe around two hours to translate it before. But now I can translate the same script within fifteen minutes. Then I preferred working on it as these technological tools actually saved an hour and twenty-five minutes.’ ‘It may be helpful that I can complete the work before my time. Then in the case of translation where it saves a lot of time.’

	Linguistic barrier	‘There is a huge lack of Bangla AI, which is another reason for not utilizing AI. Because if you want to search, check something or even make news, you need to have those information. Also, if you even put some information, AI does not have that capability to produce a proper news in Bangla.’
Dual perspectives	Optimistic and encouraging about AI	‘Everyone says that they will lose jobs. But it will be kind of opportunity actually. It might make the job a lot easier. It is a big scope for journalists’ capacity building, removing many kinds of hassles. I think it will create opportunities in the future. Besides, I don't think it will be a big challenge. I am undoubtedly positive! Because I can see positive aspects. It seems to me no problem.’ ‘Implementing AI is the third wave that we have been going through. We have already faced two other technological waves including digital media and social media inclusion before. So, there is no way to avoid AI in future, matter is how much prepared we are and how to organize it with different policies. I would say policies must be taken from the organization and government initiatives.’
	Employment uncertainties	‘I see it positively, but I personally am not in favor of using it on a large scale because especially in a small country like ours and where there is a lot of poverty, the employment of people is very important. If AI is implemented, many people will lose their jobs. Job security in the media of Bangladesh is already low.’ ‘We can see that there is already an economic pressure in Bangladeshi journalism. Not all organizations are able to pay the salaries and allowances. Ah, the cost of living has also increased. So, where 40 to 50 people used to run

		an organization, now the owners are trying to run it with 20 people. So, there is a risk that many will lean towards machine rather than human journalists because of the labor and time and many benefits will be available here.’
	Human, Vs machine	‘AI presenter can never give the same expression and the appeal of human never diminishes. Emotion is definitely a big part when a presenter reads the news, he is not only reading the information, s/he is serving the emotion or feeling inside the news and through that the audience connects with the news.’ ‘In these cases, AI never can replace human, at least a human will be needed always even though AI gets more updated. There is a room for individual-to-individual understanding, no matter how much you train AI, it will never be able to do certain tasks of a human journalist.’ ‘Some of the machines have limitations too, the limitations are everywhere, like in many cases, to collect specific information, I have to maintain a good relationship with my sources...in order to get information from that source. Any journalist has to maintain that relationship based on human emotions. One must maintain those emotions or talk to the source with a smile. AI will have a deficiency in building a good relationship.’
	A double-edged sword	‘Yes! As a journalist I have some number of concerns. There is a place such as AI and its tools can’t ensure journalistic ethics and its moral background properly. The biggest challenge will be to prevent the misleads, fake news, though nowadays AI is becoming stronger to find out and prevent these fake news by itself but technology in Bangladesh is not that developed yet.’ ‘Like some subtle meanings of different words like Massacre, Genocide both have different meanings, so what I am indicating in Bengali,

		that's sensitive. As you know, 1971 is a very sensitive issue in Bangladesh. If subtle meanings of different words like massacre, genocide replaces, it will be a big ah mess, maybe even vandalism, maybe a lawsuit against the journalist or the whole media organization.'
Roadblocks to AI	Tech Tensions	'There is a negative approach towards AI. It is our mentality of taking the utility of AI tools in a positive way has not yet developed in these newsrooms.'
	Inadequate expertise and effective training	'Institutionally no opportunity has been taken for particular training on AI or its any tools because a huge absence of experts, experts are yet to be found.' 'There is a big need in building skilled manpower. Since I work in the top media of Bangladesh, if skilled experts are missing here then I think there is little chance of including experts elsewhere.' 'One of the reasons is financial, because when you will implement AI, it will not be free of cost. Newsrooms have to pay certain amount for subscription. When these newsrooms are unable to pay their employees, integrating AI would be a challenge.'
	Linguistic barrier alias dominance	'There is a huge lack of Bangla AI, which is another reason for not utilizing AI. Because if you want to search, check something or even make news, you need to have those information. Also, if you even put some information, AI does not have that capability to produce a proper news in Bangla.' 'Various obstacles in this regard because the languages that we have in Bangladesh and South Asia such as Bengali and Hindi even Arabic, have

		not yet been developed with high-powered AI. This is an obstacle for our region.' 'Who will build Bangla AI, input the information with proper meanings, that's the question. It might take long years, it is a matter of ignorance I would say.'
Potentials, and possibilities of future application		'We are running a pilot project. <i>Basically, it is a website we bought from an Indian company. It's a content management system.</i> it is experimental phase though and we have been running it for a few days.' 'We are uploading news by processing ourselves, then we are observing if this tool can make headlines properly? It was a news of 800 or 200 words. Yes! it is experimental, very experimental though.'

6.1.1 Very limited, personal, and experimental use

The utilization and uptake of AI are still very low and limited to personal-level interests. Overall, this theme answers two overarching questions (RQ:1) and (RQ:2) answering how the participants perceive both the usefulness and ease of use of AI. Here, every participant in this study shared that they are using this technology out of their personal interest, as most of their news organizations have neither taken any formal initiative nor established basic policies regarding AI and its tools. While analyzing the interviews, each reporter, sub-editors, and editor offered phrases like "Very limited", "Low", and "Nearly no usage" to describe the current integration of AI. Although these respondents and their co-workers are using and experimenting with particular AI tools like ChatGPT, Google Translator, and Google Bard (now known as Gemini). This usage is limited and can't be addressed directly related to the news-making

process, but it's more related to checking and searching background information. Thus, it answers research question 3, regarding how participants define AI and its role in their newsroom tasks. The findings reveal that journalists and editors currently understand AI and are using it in a limited capacity, primarily for checking and searching background information, rather than directly in the news production process.

‘Online and Television media are using AI in an experimental basis also for some time, but it is not fully started from organizational level. Journalists are also using AI in their newsrooms in their own personal interest...But actually, not any media house has started it institutionally. I have talked to many journalists who know me on a personal level, working at technology beats even, they are also trying to use it in their personal level,’ (Participant 2).

Thus, the participants’ perception regarding the usefulness of AI is discussed as “Task-specific utilization” which indicates that the participants in this study along with their co-workers are using AI tools in a very limited manner, only for some specific tasks. The participant from Dhaka Post said, ‘We are using AI to translate basically conventional Bengali to English, ChatGPT is used the most for this task. To check whether the translation is correct and then whether there is any grammar or sentence problem, basically it is for writing and correcting. In English media of Bangladesh, in many cases, AI tools are being used for news editing also...Like from my own personal experience, I use it by giving some hints or a few information, it creates pictures according to my information, Sometimes, I am giving texts, AI tools making video according to my text,’ (Participant 2).

A senior editor from Bangla Tribune told, ‘When I look for something on the computer, or because I can't figure out something in particular, I take help from AI, like we often go to Google to find something. I try to write a few words, before making a news then AI suggests me

the background or some relevant materials. Then sometimes to translate, like I upload the Bengali and it translates into English. I would say it is not a total use or reliance, but it is probably a little modify level that I want to go through before writing a report,’ (Participant 4).

Based on the information provided by the participants, it is found that at present they are using AI (e.g. ChatGPT, Google Translator, Microsoft Translator, Copy.ai) for tasks such as background research or information checking, translation, and occasionally paraphrasing as well, ‘I did not use that much for organizational works, but I like to explore personally, like sometimes for writing, checking information to get an idea, ...Also it is a good source that can be used for paraphrasing and getting news updates,’ (Participant 6).

From these interviews, it is evident that significant AI tools have been used for graphics, image generation, and converting voice to text, rather than for text-related tasks or content distribution. Therefore, the limited use of AI is more prevalent in mainstream online portals and Television media compared to newspapers, as a participant from Prothom Alo shared,

‘We use graphics a lot in the case of video editing, then in the case of image generation, selection, my colleagues are utilizing ChatGPT in a good way. It also an impact that already digital transformation has been occurring in Bangladesh,’ (Participant 8).

A senior reporter from Jamuna Television shared his experience of working with AI, ‘I have used AI in my reporting in a limited way...But in the case of graphics design, I may have taken the help of AI tools, especially in the case of locating a specific place to show it virtually in the news. It has actually been done on a small scale though,’ (Participant 12). Also, on online or social media-based news platforms,

‘Those who are creating various news content through social media in Bangladesh, are using a lot of AI tools. Like they are using Canvas software. They are using some of it for video editing, where they input the footage on any topic and the software will automatically make the content,’ shared a special correspondent from Channel 24, (Participant 6).

The only sub-theme that has been categorized under this theme is, “*Selective application for presentation*”. Channel 24 is the first broadcast channel in Bangladesh to start incorporating AI into their news presentations. They have continued to feature the AI-generate female presenter Aparajita in one of their technological news programs. Even though Channel 24 incorporated an AI female presenter, Aparajita into their news presentation and one weekly regular show, till now Aparajita only performs through voice-over. A dedicated producer and editor still handle the fundamental tasks, from preparing content to uploading it to software,

‘For now, the AI-generated presenter is only doing voice-over. We have a dedicated producer for this task basically. There is also an editor for the program who is in charge of control of producing this program and he prepares the content by himself. After preparing the content with information, and scripts in Bengali, he finally uploaded it to a software. Then the AI presenter whose name we have given as Aparajita, reads the news or presents the information,’ (Participant 6).

What are the perception and acceptance levels of this AI-generated news presenter among the mainstream media and audience? This participant from Channel 24 shared his views, ‘When we introduced Aparajita, we received a lot of negative reactions in social media because the audience could not find the space for emotion and empathy in the way of news presentation,’ (Participant 6).

From the perspective of this respondent, it is salient that an absence of emotions which is one of the prominent elements of news, leading the audience's lack of appreciation for the AI presenter. Another participant from an online news portal shared the perceptions, emphasizing how the AI presenter is operating, 'Well, here is misconception again. How would you define news making process? Yes, Channel 24 has done a big branding by integrating AI presenter in their news presentation, but there are man-forces who are working behind this. It does not work like a basic AI tool where it creates content and distributes by itself too. Rather it lacks any kind of emotion, tones that a human presenter would have,' (Participant 2).

While other selected news organizations have not started any organizational initiatives for AI, Channel 24 serves as one of the present examples where practices such as news and weekly show presentations are implemented. This demonstrates a limited but at the same time primary initiative of integrating AI technology and its usefulness in Bangladeshi mainstream newsrooms.

Generally, news production comprises three primary processes including research or background information checking, production, and finally distribution (Belair-Gagnon & Revers, 2018, Munoriyarwa et al., 2021). From the present context of Bangladesh, while AI tools are not extensively used for news production and distribution, these tools are being employed at a preliminary level for tasks including research, background information checking, and news presentation. However, the adoption level remains notably low and very limited in the mainstream newsroom, with only one of the selected media organizations in this study.

6.1.2 Absence of ease

The second primary research question centered on the ease of use of AI. Following the theoretical framework, during the one-to-one interviews, the conversations emphasized each participant's perception and experiences with the ease of integrating AI into their daily work. These findings relate to specifically research question 2, as it focused on the second determinant PEU of the TAM. Participants shared that due to a lack of proper AI knowledge, uncertainty about its practical implementation, and limited professionalism, the ease of use cannot be effectively analyzed. Rather, the theme "Ease absence" emerged in this analysis phase, as AI usage is primarily at personal and experimental levels instead of integrating at an organizational level.

'Because the topic of artificial intelligence is new in Bangladeshi newsrooms and it's not that long since we have started talking about this, so those of us who actually work professionally are still at the theoretical level. Practically, it has not yet been possible to work because it has not reached a very good professional level,' (Participant 4). Another journalist from DBC News Channel added, 'Well, what I think is my perspective about AI in Bangladesh, which is still used in a very limited way in the newsrooms, as far as I know, only one news in Bangladesh uses this technology that is Channel 24 which is limited too. Only in the case of news presentation, it is not used for any news gathering and writing process. There is no actual involvement or use of AI in these news-making processes,' (Participant 11).

'Actually, the matter is very new in Bangladesh, I can say that it is not that familiar. This is still very, very rare and has been used as a private initiative. If we talk about all the journalism in Bangladesh, then only one thing I can say is that it is not launched officially,' (Participant 5).

Under this theme, the researcher has identified a sub-theme “*AI as time savior*”, where the ease of AI use focused on “Reducing the time spent on”. Two of the participants from a newspaper and a news channel said, ‘It may be helpful that I can complete the work before my time, it can be helpful in reducing the time spent on background checks or any past history while making any news. I think that these are... Then in the case of translation where it saves a lot of time. Here AI tools really play a role,’ (Participant 9).

‘I believe that it can help in copy editing, rundowns, and uploading the news. Yeah, artificial intelligence can do it very quickly,’ (Participant 14).

A founder of an online news portal and current editor shared, ‘When I was working with it, it would take maybe around two hours to translate it before. But now I can translate the same script within fifteen minutes. Then I preferred working on it as these technological tools actually saved an hour and twenty-five minutes,’ (Participant 7).

It has been observed that all participants, including senior reporters, sub-editors, and editors from various media including newspapers, Television, and online news portals, reporters who have experience of five to around twenty-five years in this journalism field, unanimously shared this same positive view of AI, ‘Sometimes there are so many information and actually the internet is crowded with thousands of websites, and it becomes difficult to find specific facts. In that case, if there is AI, it can bring me all the information in a moment, it is a very useful thing. In cases of script writing, and audio-visuals, I would emphasize visuals where AI tools can do the editing and create visuals in a short period...The work will be done very quickly for us and now sometimes it is a very time-consuming matter. Now it appears that I have to manually write a script and then edit it on the panel. It takes about two to three hours to do everything. If there is

AI involvement, I believe it will complete the whole process within at least an hour,’ (Participant 11).

6.1.3 Dual perspectives

This theme prominently answers RQs 3, 4, and 6(a), 6(b) as it addresses the dilemmas and uncertainties surrounding the use of machines instead of humans, as well as the psychological apprehension about using AI tools.

6.1.3.1 Optimistic and encouraging about AI: All the respondents have been positive about the integration of AI in newsrooms and have expressed a welcoming attitude towards future changes in journalism. It directly relates to the determinant perceived facilitation under PEU, which refers to the factors that highlight the ease and benefits of integrating AI (Venkatesh, 2000; Venkatesh & Bala, 2008). While journalists have expressed positivity about AI helping in their daily work and saving time, editors have shared their optimism about advancing in the competitive global journalism landscape. Prominently, it has been observed that all participants recognize and are ready to accept the rapid pace of technological change. Having already faced and adapted to various technological shifts, they maintain a positive attitude toward any new advancements. It is clear to survive and stay competitive in global journalism, they must continually prepare for and embrace these changes and challenges.

Editor of the broadcast channel-based online media platform shared, ‘Implementing AI is the third wave that we have been going through. We have already faced two other technological waves including digital media and social media inclusion before. So, there is no way to avoid AI in the future, matter is how much prepared we are and how to organize it with different policies. I would say policies must be taken from the organization and government

initiatives,’ (Participant 3). The founder and current editor of the online news portal also said, ‘Those who are thinking about job loss, there is nothing to lose! When we started online journalism, many people said that online journalism is not going to work out and what are the necessities of it. But that didn’t even take us a decade just to change their whole perception. The same will happen with the AI. Those who will actually lose their job are those who cannot understand or accept this,’ (Participant 7).

He also referred to the changed situation that a developing country like Bangladesh faced, ‘Once there were villagers who could not even use mobile phones, now they are using the Internet in their phones. Once journalists in the city opposed typing on computer as they were so comfortable using the typewriter machines but look at the newsrooms now. Everyone is using the Internet and typing on the keyboards. That’s how change takes place in a society. Thus, the time has come to take things positively,’ (Participant 7).

Another participant who is working as an executive director of a broadcast channel said, ‘Everyone says that they will lose jobs. But it will be a kind of opportunity actually. It might make the job a lot easier. Yes, there is a place for job cuts in some cases. But it is a big scope for journalists’ capacity building, removing many kinds of hassles. I think it will create opportunities in the future. Besides, I don’t think it will be a big challenge. I am undoubtedly positive! Because I can see the positive aspects. It seems to me no problem,’ (Participant 15).

Now, if we focus on the shared perceptions of the reporters, sub-editors:

‘Just as there is a place for fear, there is also a place for possibility. Though I have mixed perspectives, like here in Bangladesh, risk owners and editors will lean towards machines rather than human journalists which will cut some people’s jobs, ... I have seen many journalists who

were not interested to learn typing in a computer before, they were literally forced to learn it. We faced these changes already, people are now doing mobile journalism, so in the future, there is no way to avoid AI,' (Participant 4).

A special correspondent from Jamuna Television said, 'Well, technology will take me forward. Now the thing is, how will I use it? If I say about the creative place in journalism, AI will create some opportunities too. Yes, there is a scope to lose a job, but I think if I stay updated with technology, focus on new projects and changes then surely yes, it will be an opportunity,' (Participant 1).

6.1.3.2 Employment uncertainties: Three participants who have extensive experience in the journalism field, directly stated their opposition to the integration of AI in their newsrooms, mentioning the concerns primarily about the uncertainty surrounding their job security. Journalists from different countries expressed their apprehensions about potential job loss (Munoriyarwa et al., 2021; Jamil, 2019). It is noteworthy that the statements of the participants in this study did not articulate fear. Rather it is more characterized as uncertainty related to their job security.

'I see it positively, but I personally am not in favor of using it on a large scale because especially in a small country like ours and where there is a lot of poverty, the employment of people is very important. If AI is implemented, many people will lose their jobs. Job security in the media of Bangladesh is already low. Umm... and there is AI and if the owners are in the position of control, one can understand that with artificial intelligence, half of the manpower can be deducted easily then maybe they will slowly move towards that. Many televisions will drop out because there is not so much investment, they will drop slowly and eventually,' (Participant 6).

A senior reporter from Prothom Alo added, ‘Oh yes! If AI can be enriched, it will be helpful on the one hand, but on the other hand, it will also create some risk. Once, there used to be a telephone from which the news from various agencies would be shared and then printed with the help of a teleprinter. And finally we the journalists used to send the report to those who are outside. Then the era of composing came and eventually it was lost too. The same will happen with the AI like several people lost their jobs when e-mail and Internet facilities became available,’ (Participant 9).

Participant 4 shared, ‘We can see that there is already an economic pressure in Bangladeshi journalism. Not all organizations are able to pay the salaries and allowances. Ah, the cost of living has also increased. So, where 40 to 50 people used to run an organization, now the owners are trying to run it with 20 people. So, there is a risk that many will lean towards machine rather than human journalists because of the labor and time and many benefits will be available here.’

6.1.3.3 Human, Vs machine: Relating to the elements of excellence and the demonstrability of results under PEU of the TAM, this theme emphasizes the perceptions of human and AI tools. Despite the prevailing uncertainty, all respondents asserted that regardless of the advancements in AI, the field of news and journalism remains one where humans can never be replaced. They emphasized the indispensable role of humans in aspects such as conveying emotions, displaying empathy, maintaining sources, making interpretative reporting, and sensitive news stories. The senior reporter from Jamuna TV shared, ‘It seems to me that this is a completely unreasonable idea that if we use this product on a large scale then our reporters, presenters or desk journalists will lose their jobs. I don't even think, if it's twenty years later or fifty years later, AI presenters can give the same expression, and the appeal of humans never

diminishes. Emotion is definitely a big part when a presenter reads the news, he is not only reading the information, s/he is serving the emotion or feeling inside the news, and through that the audience connects with the news,' (Participant 12).

'In these cases, AI never can replace humans, at least a human will be needed always even though AI gets more updated. There is room for individual-to-individual understanding, no matter how much you train AI, it will never be able to do certain tasks of a human journalist, it is my opinion. Journalism is a profession where you not only have to maintain your moral ground but also have to make an impact. With your own objectivity, it's a very matter of empathy. Like, if you are dealing with a criminal case, then as a journalist you have to listen to the criminal, you also have to listen to the statement of the innocent, and you have to confirm the position of all sides from your own human perception. No matter how much AI is developed some places still need empathy, emotions, and specific tones for making a report. I can't say that I am 100% sure, but I mean I am right, I am not convinced that it is possible for AI to make editorial decisions like human journalists,' (Participant 8).

Emotions in news also include news commentaries, opinion pieces, and editorials where journalists share their own perspectives (Feldman, 2011; Munoriyarwa et al., 2021). These pieces have a particular feature and role in different public spheres like politics. It creates a scope of critical discussions where journalists also can play a crucial role in shaping an ideal society within their professional domain. Focusing on these elements of news, a reporter from Somoy TV shared his perceptions, 'Some of the machines have limitations too, the limitations are everywhere, like in many cases; to collect specific information, I have to maintain a good relationship with my sources...in order to get information from that source. Any journalist has to maintain that relationship based on human emotions. One must maintain those emotions or talk

to the source with a smile. AI will have a deficiency in building a good relationship. It cannot maintain a source like a government official, a victim,’ (Participant 14).

6.1.3.4 A double-edged sword: While the editors and journalists expressed optimism and highlighted potential possibilities, the participants also focused on their confusion and doubts about the dilemmas that AI could present.

A participant from Prothom Alo shared, ‘Yes! As a journalist, I have a number of concerns. There is a place such as AI and its tools can’t ensure journalistic ethics and its moral background properly,’ (Participant 8). She also added, ‘The biggest challenge will be to prevent the misleads, fake news, though nowadays AI is becoming stronger to find out and prevent this fake news by itself technology in Bangladesh is not that developed yet.’ Newsroom editor from DBC News said, ‘News written and processed by artificial intelligence, can we rely on one hundred percent? Sometimes we have to deal with sensitive information, there might be chances that the news is not one hundred percent authentic. We have to cross-check it, and I think completely AI dependency is not really possible,’ (Participant 11).

Respondents particularly indicated the issue of the sensitivity of the information and importance of cross-check. This is significantly relevant in the context of present Bangladesh, where journalists easily face legal actions under various laws and draconian acts, leading to significant harassment for themselves.

‘Let's say about Google Bengali Translator. We need the proper meaning of every word to get published in the newspaper, if AI changes any word, it will create different meanings. Yes! It will be a sensitive issue. Like some subtle meanings of different words like Massacre, Genocide both have different meanings, so what I am indicating in Bengali, is sensitive. As you

know, 1971 is a very sensitive issue in Bangladesh. *If subtle meanings of different words like massacre, and genocide are replaced, it will be a big ah mess, maybe even vandalism, maybe a lawsuit against the journalist or the whole media organization,*' (Participant 10). The senior editor from Bangla Tribune added, 'Then there are a lot of obstacles like I can't even rely on AI for some sensitive issues even if I want to. There is a fear of it. If things are not cross-checked then it can be a disaster and it will create a threat to our safety and our normal life,' (Participant 4).

6.1.4 Roadblocks to AI

Roadblocks to AI factors including technological apprehension, lack of support, linguistic, and cultural barriers mainly affect the PEU, as these influences make technology harder to use. These challenges may lead to increased resistance to adopting AI, as participants in this study shared their perception of AI tools as less accessible.

6.1.4.1 Technological Apprehension/ Tech Tensions: Though the participants expressed their optimism, there is a prevailing fear of technological apprehension within their current media organizations. Several reasons contribute to this existing fear of AI. In regions such as Bangladesh, there is a cultural tendency to avoid new technological changes primarily. People here are generally not receptive to new technological waves, as observed during the COVID-19 pandemic when many activities shifted online. Participant-4 from Bangla Tribune said, 'The practice has not yet started in Bangladesh, and we probably don't take it very seriously'.

'Yes! They have been in journalism for a long time, but they are a little far away from technology. Hmm, there is a kind of fear among them about this. They want to stay away but at

the individual level, some are personally updated,’ (Participant 5). The editor of an online news portal said, ‘There are two kinds of people in our society, one is the kind of people who think that this technology will limit their way,’ (Participant 7). The Special correspondent from Channel 24 added, ‘There is a negative approach towards AI. It is our mentality of taking the utility of AI tools in a positive way has not yet developed in these newsrooms,’ (Participant 6).

6.1.4.2 Inadequate expertise and effective training: One of the prominent roadblocks to AI is the absence of proper knowledge, skilled expertise, and effective training. The senior editor from Bangla Tribune said, ‘Institutionally no opportunity has been taken for particular training on AI or its any tools because a huge absence of experts, experts are yet to be found,’ (Participant 4). An editor shared, ‘Reasons are that the media houses are lagging behind in terms of technology when developing countries are moving forward. As media organizations are not technologically sound employees are not prepared too. An absence of knowledge is prevailing in Bangladesh,’ (Participant 5). The senior sub-editor from Prothom Alo added in this study, ‘There is a big need in building skilled manpower. Since I work in the top media of Bangladesh, if skilled experts are missing here then I think there is little chance of including experts elsewhere,’ (Participant 10).

Participants also indicated significant reasons behind these absences. First, financial constraints and budget limitations that most media houses are currently going through,

‘Financial crisis and the limited income in our newsrooms are common here. As a result of this economic crisis, a lack of desire can be seen among journalists. Because we already know that with less money we cannot do well with updated technology,’ (Participant 4).

‘One of the reasons is financial because when you implement AI, it will not be free of cost. Newsrooms have to pay a certain amount for subscriptions. When these newsrooms are unable to pay their employees, integrating AI would be a challenge,’ (Participant 11). An editor of a broadcast channel shared, ‘Here budget, advertisement market is not doing well. Overall, mass media could not become proper media for mass people,’ (Participant 3).

Second, the apathy of owners and policymakers, ‘Most media organizations do not have an editorial policy yet, how can they create any policy on AI when the proper knowledge is absent? They may say things orally but most of the organizations do not have any written policy which is more common in television channels,” (Participant 12).

A reporter from Somoy TV who is enthusiastic about technology stated that ‘The media industry is actually not stable so far what is meant by professionalism. It is moving forward in an amateur way. So, in Bangladesh, the media houses, especially television, and newspapers, will be facing big loopholes when it comes to using AI,” (Participant 14).

The third and subtle factor also indicates the lack of proper courses in departments and universities which is a significant issue. AI is still relatively a new arena in Bangladesh, with limited work research initiated so far. Even for subjects like data journalism, which are not entirely new, many media and journalism departments in universities have not fully implemented them. This raises a critical question: do our universities have the necessary resources and expertise to offer updated courses and provide practical knowledge in these emerging fields?

The senior sub-editor from Prothom Alo added, ‘I don't think anyone has included AI as a new course in the universities. So, it's a big gap. Let me give you an example. When the concept of mobile journalism first came to Bangladesh, nobody knew about that. Then mobile

journalism was introduced as an idea and the world of ah online was changing so fast. Basically, it took time to implement this new concept,’ (Participant 10).

Lastly, another factor contributing to the roadblock is the audience’s pessimistic perspective. Two participants from broadcast media including one from Channel 24, the very first media to introduce an AI presenter in their news and weekly technological show shared their perspective on this, ‘Most of the audience said that not this presenter is just reading but human expression is evidently missing. The audience is not getting that feeling,’ (Participant 6). Another senior reporter from Jamuna TV said, ‘Once channel 24 launched AI presenter, there were various criticisms from the audience,’ (Participant 12).

6.1.4.3 Linguistic barrier alias dominance: One of the main remaining challenges is the lack of Bengali AI and the significant absence of authentic Bengali information with accurate meanings. All participants in these interviews particularly emphasized this absence, highlighting a language barrier on a global scale, ‘There is a huge lack of Bangla AI, which is another reason for not utilizing AI. Because if you want to search, check something, or even make news, you need to have that information. Also, if you even put some information, AI does not have the capability to produce proper news in Bangla. Media organizations mostly depend on this Bangla language, right? So, what will they find? What will the journalists produce? *It indicates language dominance too. It is a matter of world market where Bengali is not that prominent,*’ (Participant 8).

A participant from the Dhaka Post expressed, ‘Various obstacles in this regard because the languages that we have in Bangladesh and South Asia such as Bengali and Hindi even Arabic, have not yet been developed with high-powered AI. This is an obstacle for our region,’ (Participant 2). The senior reporter from Prothom Alo said, ‘In English, you get support easily

but if you consider Bangla, you will not get that in terms of technological tools. I would say it is a domination from overall the global market,’ (Participant 9).

Is it the only matter of dominance or there are other reasons too? The founder and editor of the online news portal emphasized that it’s because of our lack of technological skills and indifference, ‘*Who will build Bangla AI, and input the information with proper meanings, that’s the question. It might take long years, it is a matter of ignorance I would say,*’ (Participant 7).

It’s same for the English newspapers because the basic information and content are in Bangla. As an editor of a newspaper said, ‘There may be a problem in terms of contents in Bengali. AI is still not able to catch up with the meaning...What do you think about English newspapers? It’s all the same because the meaning and information are absent there. In that case, too not any media can be completely dependent on AI,’ (Participant 5).

6.1.5 Potentials and possibilities of future application

Despite the current limitations and the absence of AI in professional and organizational contexts, two media organizations except Channel 24 selected in this study, have begun pilot projects and initiatives. They are collaborating with experts from institutions (e.g. BUET) and exploring the implementation of AI software developed in foreign countries. In this context, participants, particularly editors and senior reporters with over ten years of experience in journalism, have shared greater optimism compared to other participants.

“We are running a pilot project. *Basically, it is a website we bought from an Indian company. It’s a content management system.* It is an experimental phase though and we have been running it for a few days”, Participant 10, from a newspaper added. What types of work they are experimenting with in this pilot project? This participant also added, ‘We are uploading

news by processing ourselves, then we are observing if this tool can make headlines properly. It was a news of 800 or 200 words. Yes! It is experimental, very experimental though,' (Participant 10).

Another participant who is an editor of a TV channel-based digital platform said, 'We have a project on our own, we will start working with the website imported from another country. Maybe by next year we will start working with it more organizationally,' (Participant 3). Another editor from a broadcast channel shared, 'We are in the process, and we *are also meeting with the experts who work in Bangladesh to see what we can utilize more. These experts are basically from the organization BUET who have knowledge in computer science and technology.* Also, we are sitting with the reporters, and we are trying to get the basic understanding of how to integrate the AI in newsrooms,' (Participant 15)

Other selected newsrooms are also at the phase to start their initiative levels, like, 'There is a discussion within us that how the AI tools will actually work, how can we help the newsroom. If AI is launched, will journalists lose their jobs or those who are doing journalism will use this? Whether it is possible to bring more dynamics to the flow of news, *it is discussed at the individual level,*' (Participant 10).

Although journalists, particularly senior journalists, and editors, along with most participants, expressed optimism about potential opportunities and training initiatives for their employees, it appears that the current situation in these newsrooms remains very basic or even less developed. Also, since this analysis is based solely on one-to-one interviews rather than on-site visits and observations, there is a possibility that participants expressed more optimism than the real situation. For instance, one editor mentioned that they would begin experimenting with AI within the next year solely, but given the current situation of newsrooms in Bangladesh, it

may take significantly longer than this estimate. Evidently, training on AI and its tools is still lacking. While training on data journalism and other journalistic skills is available, participants in this study noted that AI training remains largely absent due to a shortage of experts in the field. Prominent reasons are lack of financial stability and proper knowledge which have been already discussed. For these reasons, it is not feasible to hire tech experts from other countries. More importantly, media organizations have yet to grasp the basics of AI and its functions. This analysis reveals a pervasive lack of understanding and preparedness in the newsrooms.

One senior editor of an online news portal shared her opinion on AI training, ‘*Institutionally no opportunity* has been taken to show that particular training on it because the winning source person is yet to be found. I hope that I will talk about this issue even if it is in the office *on a personal initiative* that we can actually use it positively,’ (Participant 4). Another respondent shared, ‘No, there is no training or module or any such encouragement or discuss or any steps to use this from our organization. Our house can be said to be very ignorant in this related matter,’ (Participant 14). Participant 11, from a broadcast news channel, said, ‘I will not say that this kind of interest is there in the newsrooms. No! So far, we (the media organization) have not taken any kind of workshop, training, or any initiative to teach in this regard.’

Given the current circumstances, the lack of training on AI and its tools, the shortage of experts, and the limited understanding of AI fundamentals among media organizations all contribute to its delay. Thus, the widespread absence of the necessary infrastructure and knowledge indicates that it will take a considerable amount of time before AI can be effectively implemented in these newsrooms,

‘What I think is in Bangladesh, before the print media, Television will start their utilization. But I think professionally in every media platform, it will take a long time. Journalists

and media organizations are not equipped in that way’, (Participant 5). Another senior reporter from Prothom Alo added, ‘So far there is no such policy. Office is not doing it and not encouraging it officially but saying employees that yes, you can do it to save time,’ (Participant 9).

CHAPTER 7. DISCUSSION

The key themes highlight that AI is currently in a stage of very limited, and experimental use with significant roadblocks. These create a double-edged sword, where the challenges coexist with significant potential for future applications. Despite these roadblocks, the possibilities for AI in journalism in Bangladesh hold possibilities offering some risks also.

This study particularly answers four questions: (i) The understanding and perceptions of the journalists and editors; (ii) What is the current scenario of newsrooms in Bangladesh integrating AI; (iii) The roadblocks behind the present situation; and (iv) Potential possibilities of AI. First, this research deepens our understanding basically by examining and analyzing, how journalists and editors as participants perceive the usefulness (PU) of AI and its ease of use (PEU) in their newsroom practices. Based on qualitative interview analysis offers new knowledge and insights that the concept of AI, its usefulness, and how to integrate it with proper skills and knowledge in the making and distribution process, which has not been started in newsrooms yet. This obviously shows a contrasting scenario where newsrooms in the developed countries, already have started this practice as one of their features in the news production. Though Channel24 has launched its news presentation and a show from an initial phase, a leading newspaper with a major number of audience, has started a pilot project, other selected newsrooms have not started or even thought of AI from the organizational approach. Under the sub-theme "Human vs. Machine," participants from organizations other than Channel24 expressed that the AI news presenter lacks the basics that a human journalist should have. They questioned whether it truly represents an AI application, given the absence of audience connection, natural pacing, and the involvement of a human team behind the scenes which aligns

with the determinants of PU, particularly in terms of demonstrability of results and output excellence. They noted that the audience does not seem to take this AI-powered presenter seriously. On the other hand, both the management and employees of Channel24 view their AI application positively, considering it a new beginning in the third wave of AI which emphasizes the relevance of the concurrent playfulness, another determinant of the PEU. Participants expressed optimism, noting that their organization has consistently been at the forefront of innovation compared to other media outlets.

Second, beyond the news organizations, if we focus on the perceptions of our participants, which is the main aim of this study, we can understand optimism and a strong non-avoidance approach among them. This indicates a positive aspect that will lead to an up-to-date newsroom equipped with modern technology, where journalism here will begin utilizing AI tools effectively. Participants have expressed their interest in AI and its applications at their individual level. These positive perceptions align with perceived usability, which is a key element of PEU (Venkatesh & Bala, 2008; Wan & Jiang, 2023). A notable pattern emerged during the in-depth interviews, participants in higher authority or management positions tended to express more positive perceptions about implementing AI in their news organizations, while journalists were generally more cautious and skeptical (two among fifteen) about its application. Like, executive directors, editors, and owners of online news portals were notably positive about AI. They referenced news organizations in developed countries, highlighting how AI has made journalism more efficient there. They expressed hope that similar advancements would benefit journalists in Bangladesh, cautioning that those who fail to adapt to evolving technologies may face challenges in the future. On the other hand, a senior journalist from the same Channel24 expressed concern about integrating AI, fearing it would lead to job losses for journalists and primarily benefit the

owners by reducing costs. This suggests a clear divide in perceptions between management and working journalists. The data indicates that both groups see AI differently, with management focusing on the potential for cost reduction, while journalists are more concerned about the impact on their job security. According to TAM, these varying perceptions are influenced by an organization's economic situation, work practices, and ability to adapt to a particular technology (Janneck, 2009; Munoriyarwa et al., 2021). Here, the economic factor plays a major role in creating contrasting perceptions between editors and journalists. Editors, who are involved in policymaking are more likely to highlight the positive aspects of their organization adopting new technology, while the actual situation might be different.

Similarly, depending on the organizations' tendency to adopt new technology, participants from newspapers and online platforms shared a more positive approach toward AI compared to those from television channels. This difference may stem from the fact that television channels lack the organized policies that newspapers have established. In contrast, online news portals tend to foster a more digital culture, encouraging a greater inclination toward technology. Nonetheless, despite these distinctions, all three types of media share similar perceptions of AI in contemporary journalism.

Third, on the flip side, this positivity leads us toward the current challenges and dilemmas that have been addressed as the roadblock of AI. Findings confirm those obstacles including finances, lack of experts to train and teach about the AI tools, and most importantly absence of Bengali AI. The language obstacle stands out prominently, with every participant strongly emphasizing their concern regarding this issue. The important question remains: why hasn't Bengali AI developed yet? Is this only with Bengali, or with some other languages like Arabic, Hindi, or Latin? Certainly, absence of initiatives, technological applications are missing to

integrate Bengali AI with proper meanings and correct information. But from the interviews, it also indicates a *dominance* significant language/s over these other languages which is not new. Journalists often struggle to find accurate and reliable information to verify or background check news stories. AI is not yet equipped to provide information in Bangla, leading to concerns among participants about the lack of accessible and meaningful information in the language.

Finally, despite these challenges and shortcomings, potential possibilities are also there. Like, a selected newspaper has started its pilot project, broadcast channels are trying to be in this competition when Channel24 has already integrated AI (though in the initial phase), and media organizations communicating with experts efficiently in this field. Overall, it suggests that journalists and newsrooms have acknowledged that AI will play a role in the future, though on a limited scale. Similar to the previous waves of digitalization and the rise of social media platforms, AI will take place gradually with human journalists. Hence, the inevitability of technology cannot be ignored. Now the question remains to be answered: when and how? Even if AI comes to these newsrooms, considering the present state of journalism in Bangladesh, will these newsrooms have the capacity to train their employees, uphold professionalism, and effectively integrate AI?

Following the TAM, the study reveals a gap in AI knowledge and application, especially among mainstream news organizations, many of which have yet to incorporate AI into their operations. This suggests that, despite global advancements in AI, journalism in Bangladesh is still in the early stages of AI integration, with limited to no training and resources. By identifying these challenges and the mixed perceptions of AI's effectiveness, the findings pave the way for future scholarly exploration of how AI could be better incorporated into journalism in Bangladesh. This includes participants' varied levels of PU about AI in enhancing newsroom

efficiency, with management generally more optimistic about AI's benefits. Conversely, journalists expressed their concerns about job security, and skepticism about AI's potential, reflecting a lack of PEU. Furthermore, the roadblocks identified in the study, such as financial constraints, the absence of experts, and the lack of Bengali AI tools, also align with TAM's notion of external variables reducing both PU and PEU. However, the optimism for AI observed in this study suggests, that as journalists gain more understanding of AI, their PEU and PU may increase facilitating a wider acceptance of AI in newsrooms.

Overall, the findings reinforce the application of TAM in understanding the varying levels of technology acceptance in Bangladeshi journalism, with organizational factors, technological readiness, and individual perceptions playing a crucial role in shaping the future of AI integration.

CHAPTER 8. LIMITATIONS AND IMPLICATIONS

8.1 Limitations

This research has certain limitations as well. First, the number of selected participants. Although the data reached its saturation (since the use and implementation of AI in Bangladeshi newsrooms has not begun yet in full-fledged), a clearer picture might have emerged if more participants had been interviewed. The inclusion of more media organizations with more journalists from different beats and particularly technology experts to check their observations would provide a more comprehensive analysis. Second, one of the fundamental research questions is to examine participants' perceptions about the ease of use of AI tools. However, due to the nearly "No implementation" of AI, this segment did not yield additional insights, except for occasional use for personal interests. With inclusion of more participants from different media organizations would add a holistic finding focusing on the perception and situation about AI integration in overall journalism in Bangladesh. Due to time constraints and the reluctance of several journalists had been a matter in this study. Initially, around ten journalists expressed interest in participating in interviews, but later, for various reasons, they declined to take part which limited the inclusion of more participants from more media organizations. Another limitation is that only two female participants were recruited in this study, though at the primary phase, several female journalists were approached for the interview. This also reflects a current challenge in the industry, less representation of women in senior positions. Finally, this study did not include any radio channels, which are also a significant part of Bangladeshi media. The primary reason is that over 28 private radio channels in Bangladesh are predominantly

entertainment-focused. Additionally, the government-operated radio station, Bangladesh Betar (1971), has not yet undertaken any initiatives related to AI. The researcher had an informal discussion with one of the employees from this station regarding the matter. This could be an arena for future research where different media organizations allow for comparisons of technological adaptation across different media organizations.

8.2 Implications and Future Directions

This study on journalists' perceptions of AI use, provides newer insights and suggests important directions for future studies. Since AI has not been officially started in these newsrooms as well as other sectors yet, this study might help with its findings and literature reviews.

As one of the first studies of its kind in newsrooms in Bangladesh, the findings highlight significant roadblocks, such as the lack of technological experts appropriate academic courses, and training in the organizations, which are significant barriers to knowing about the integration of AI properly and professionally. Future research can focus on developing and assessing these programs, finding ways to overcome existing barriers, and examining how AI impacts journalism over time. Additionally, the findings can guide changes needed in journalism education by including AI literacy and practical skills, enabling future journalists to use these tools effectively in news production. Overall, this study not only adds to our understanding but also emphasizes further exploration into the role of AI in the newsrooms of Bangladesh. While other developing countries are working on establishing policies for AI in journalism, this study will provide insights, guiding the development of appropriate policies for the future use of AI in the media sector. This is one of the major strengths of this paper as it is amongst the first qualitative

research studies to examine perceptions of current journalists' about AI based on the TAM model. Though the participants were limited, these in-depth interviews, allowed us to get insights into journalists' perceptions, the roadblocks, and a few positive potentials.

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VITA

1. **Educational institutions attended and degrees awarded**

University of Kentucky

Department of Communication, College of Communication and Information

Graduate Student, Master's in Arts, Estimated December 2024

University of Dhaka

Mass Communication and Journalism

Faculty of Social Sciences

Master of Social Sciences (December 2017)

University of Dhaka

Mass Communication and Journalism

Faculty of Social Sciences

Bachelor of Social Science (2016)

2. **Professional positions held**

Assistant Professor (October 2023- Continued)

Department of Department of Mass Communication and Journalism

Faculty of Security and Strategic Studies

Bangladesh University of Professionals

Teaching Assistant (August 2022- April 2024)

University of Kentucky

Department of Communication

Lecturer. (Jan 2019- September 2023)

Department of Department of Mass Communication and Journalism

Faculty of Security and Strategic Studies

Bangladesh University of Professionals

Teaching Assistant (February 2018- September 2018)

Department of Journalism, Communication and Media Studies
State University of Bangladesh

3. **Scholastic and professional honors**

Government scholarship for graduate academic excellence, 2019

Awarded by Secondary and Higher Education Board, Bangladesh

Dr. Professor Seetara Parveen Award 2016 for Academic Excellence

Awarded by the Mass Communication & Journalism Department, University of Dhaka.

Ataus Samad Memorial Scholarship 2016 for Academic Excellence

Awarded by the Mass Communication & Journalism Department, University of Dhaka.

4. **Professional Publications**

- Parker, K., Ambegha, A., Deffendall, M., Pfeiffer, S.S., Adams, M., Clements, C., Scheuer, S., Guay, M., Abney, D., Rahman, N., & Ribott, M. A. (2023). Developing Strategies to Engage a New Generation of Donors: Fostering Relationship between Gen-Z and Non-Profit organizations. *Journal of Philanthropy and Marketing*.
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5. Nadia (Nahrin) Rahman